Analysis of Housing Prices in 45 Major Canadian Cities Using Sampling Methods

Kiu Ching Pak, Mihret Gebremedhin, Emilia Ricci

Abstract

Housing prices in Canadian cities vary significantly across different cities, which is making accurate estimations harder when using sampling methods. This study aims to use different sampling designs to estimate the population mean and total housing prices, then compare their performance. We applied different methods, such as: Simple Random Sampling (with and without replacement), Stratified Sampling (Proportional and Neyman Allocation), and Cluster Sampling (one-stage and two-stage designs) with a data set of 35,768 house listings from forty-five major cities. Then, we evaluated their performance by the mean error and variance. From the results, we can see that the Stratified Sampling - especially by Neyman Allocation - produced the lowest variance and most accurate estimates. Since there is high variability between cities, Cluster Sampling produced the highest variance, making itself the worst method for a study such as this one. These results suggested that when choosing a sampling method, accounting for variability across strata is essential, so Neyman Allocation provided the best results for this dataset.

Keywords

Clusters: subgroups of an entire population that are formed with the intention of each group being a smaller representation of the population.

Houses/Homes: referred to detached, semi-detached, and townhouses that are purchased for permanent living; any form of apartment complexes, plexes, and movable houses are not considered homes in this study due to their attached nature.

Major Canadian Cities: the most highly populated cities within Canada; working with the forty-five most populated Canadian cities.

Observational Units: the individual units in the study that have the needed information collected from.

Outlier: a unit which measurements can be taken from that vastly differs from the majority

Stratum/Strata: subpopulations of an entire population that do not overlap.

Introduction

The rapid rise and variation of housing prices across Canada has made real estate analysis an increasingly important topic for a broad spectrum of individuals. It is most concerning for those who fall within the average knowledge and exposure to housing prices,

as they may be unaware of factors affecting the fluctuation of individual homes particularly. Consequently, understanding how housing prices differ across major cities can assist in making informative financial decisions, urban planning, and economic policy. This study focuses on estimating and comparing housing prices across major Canadian cities using a variety of sampling methods taught in the MATH 3430 course. Our general interest lies in determining how different sampling techniques perform when applied to real-world, high-variability data - in this specific study, the prices of homes.

The data used in this study consists of home price listings collected from the forty-five most populated cities in Canada, obtained from a publicly available dataset on Kaggle. The full population includes 35,768 home units, which are the observational units of this study. Said observational units possess the primary variable of interest: the home values. This dataset provides a comprehensive representation of housing markets across diverse environments, allowing for meaningful comparisons between cities with varying price levels. The true population mean housing price is approximately $943,296, with a total housing value of about $33.7 billion. Said values were produced through the use of the specific lines of R codes in Figure 1.

```r
# Basic setup
N <- nrow(data)
y <- data$Price

cat("Population size N =", N, "\n")

## Population size N = 35768

cat("Number of cities =", length(unique(data$City)), "\n")

## Number of cities = 45

# True population values
pop_mean <- mean(y)
pop_total <- sum(y)
pop_variance <- var(y)
```

```r
cat("True population mean =", pop_mean, "\n")

## True population mean = 943296.3

cat("True population total =", pop_total, "\n")

## True population total = 33739820335
```

Figure 1. True Population Mean and Total Housing Value.

The primary objective of this study is to evaluate and compare the effectiveness of different sampling methods in estimating the previously defined population parameters: the population mean and total housing prices. To achieve this, several sampling techniques were applied. They are comprised of: Simple Random Sampling Without Replacement, Simple

Random Sampling With Replacement while keeping all duplicates, Simple Random Sampling With Replacement while removing all duplicates, Proportional Stratified Sampling, Neyman Stratified Sampling, One Stage Cluster Sampling, and Two Stage Cluster Sampling. We aim to assess the precision and accuracy of these methods using measures using mean error and variance. Ultimately, this analysis seeks to identify which sampling method(s) provide the most reliable estimates for housing prices when faced with substantial variability across cities.

Methodology

As previously stated, this investigation applied a vast range of sampling techniques in order to estimate the housing prices across the forty-five most populated Canadian cities. Once completed, several comparisons were made between: the population mean and sample means, the population total and mean totals, the sample mean variances, and the sample total variances. Each sample survey design was used to highlight their strengths and weaknesses when practically applied to a real-world problem faced with high variability between the geographical locations of the observational units. This allows for a comprehensive evaluation of accuracy and efficiency across different approaches.

We begin with Simple Random Sampling, as this is considered to be the fundamental probability sampling method. For Simple Random Sampling Without Replacement, units are selected randomly from the full dataset, with each round of sampling having one less unit to choose from. Consequently, this results in no duplications, ensuring equal, rising inclusion probabilities for all observations as the units are chosen for each round of random sampling. In contrast, Simple Random Sampling with Replacement allows for repeated selection of the same unit, and we consider two variations: one that retains duplicates and another that removes them to assess the impact on sample size and estimator performance. The sample size is determined using a relative tolerable error of two percent, ensuring a balance between precision and feasibility for this study.

Next, Stratified Sampling was utilized, where the population was divided by city, resulting in forty-five, non-overlapping strata. Consequently, this approach accounts for heterogeneity across cities, attempting to improve the estimation precision. Two allocation strategies were used: Proportional Allocation and Neyman Allocation. Proportional Allocation accounts for sample sizes being relative to the population of each individual strata, while Neyman Allocation further incorporates within-stratum variability to minimize overall variance. By allocating more samples to cities with higher variability in housing prices, Neyman Allocation was suspected to produce more efficient estimates.

Finally, we applied Cluster Sampling, treating each city as a cluster. In the one-stage cluster design, a subset of cities was randomly selected, and all housing units within those cities were included in the sample. In the two-stage design, a subset of cities was first selected, followed by random sampling of a fixed number of housing units within each selected city. We chose fifty units to be represented from each cluster in this study. When

thinking about applying this strategy without the use of R code, this method would have reduced data collection costs; however, that would come at the sacrifice of precision due to the variability between clusters. This will soon be shown in the results.

For all methods, we estimated the main population parameters of our study: mean and total housing prices. The performance of each sampling technique was evaluated using statistical measures such as variance and mean error, allowing a direct comparison of their accuracy and reliability to be made.

Results

The estimated population mean and total housing prices differed across the various sampling methods used. There were prominent variances in the accuracy and efficiency across the methods. The following found results were evaluated based on the mean error, variance, and overall reliability of each individual sampling method.

Theoretically, the estimates that are closer to the true population mean will signify greater accuracy. Consequently, the methods which produce estimates that greatly differ from the true population mean will indicate poor accuracy.

Figure 2. displays the mean error for each sampling method, this was found through using the provided R code. Recall, the mean error is the difference between the true population mean and the estimated mean (of each sampling method).

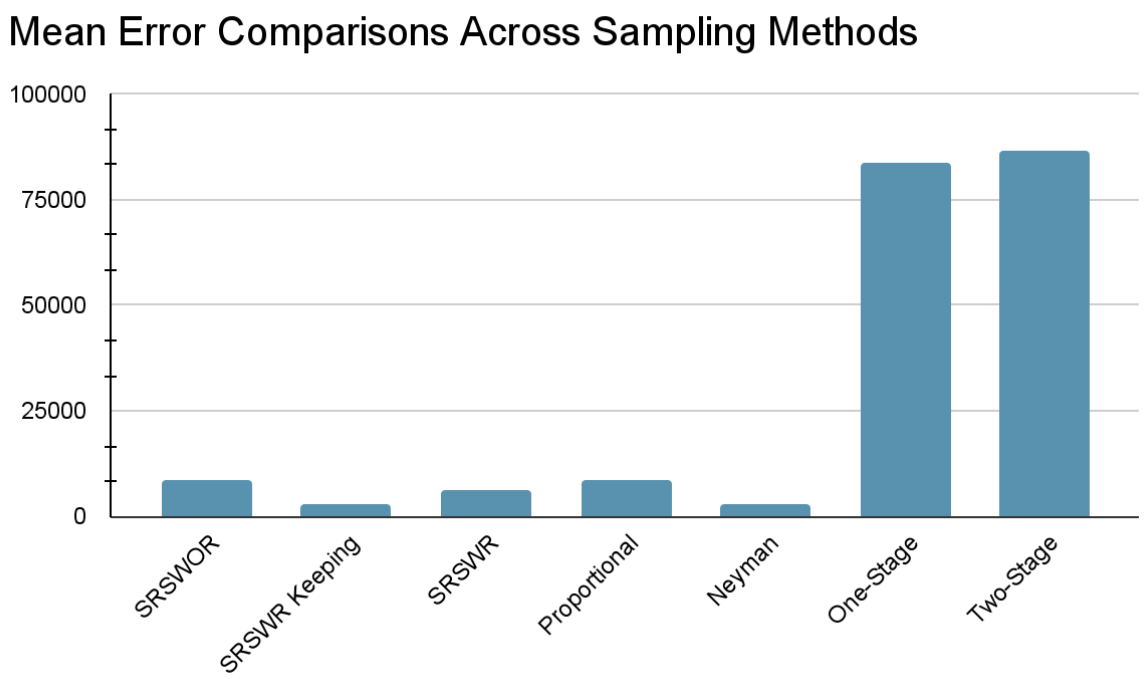

Figure 2. Mean Error Comparisons

Lower mean errors indicate that a sampling method consistently produces estimates close to the true value. The above results suggest that accounting for variability within strata significantly improves estimation accuracy. Therefore, Neyman allocation has demonstrated the best performance for the mean error. Conversely, cluster sampling performs poorly due to high variability between cities, which is not adequately captured when only a subset of clusters has been selected.

Figure 3. displays the variance of the estimated mean for each sampling method, this was found through using the provided R code.

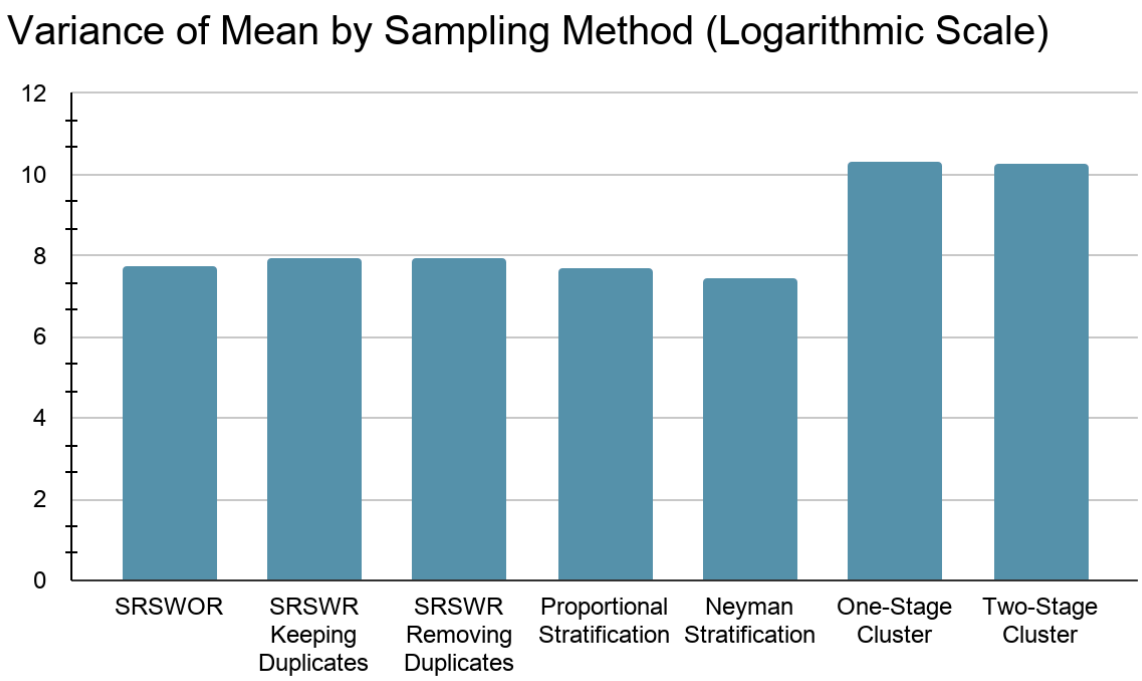

Figure 3. Variance of Mean

Variance measures the consistency of the estimates across all the samples. The lower the variance, the more stable and reliable the results will be. The superior performance of Stratified Neyman sampling reflects its ability to allocate more samples to high-variability cities, reducing overall uncertainty. In contrast, cluster sampling leads to unstable estimates, as entire cities are either included or excluded, amplifying variability.

To summarize all the data found in a neat and organised way, Table 1. Shows all key findings, and rates the overall performance of each sampling method.

Sampling Method	**Accuracy (Mean Error)**	**Precision (Variance)**	**Overall Performance**
Simple Random Sampling Without Replacement	Moderate	Moderate	Acceptable

Simple Random Sampling Keeping Duplicates	High	Moderate	Strong
Simple Random Sampling Removing Duplicates	Moderate	Moderate	Acceptable
Proportional Stratification	Moderate to High	Low to Moderate	Strong
Neyman Stratification	Very High	Very Low	Best
One-Stage Cluster Sampling	Low	High	Weak
Two-Stage Cluster Sampling	Low	Very High	Weakest

Table 1. Summary and Rating of Sampling Methods

The results deem that Neyman Stratification is the most effective method for this study, achieving both high accuracy and low variance. This is due to its ability to account for differences in variability across cities by allocating more samples to areas with greater dispersion in housing prices.

The three Simple Random Sampling methods provide reasonable estimates but do not account for underlying population structure, resulting in moderate performances.

In contrast, the Cluster Sampling methods perform poorly due to the high variability between cities. Since entire cities are either included or excluded, important information may be lost, leading to higher errors and variance. This becomes even more apparent when one of the cities could be an outlier, either due to extremely high or low housing prices. Figure 4. highlights this well, as it displays the variability of the needed housing prices in certain cities that can be considered outliers.

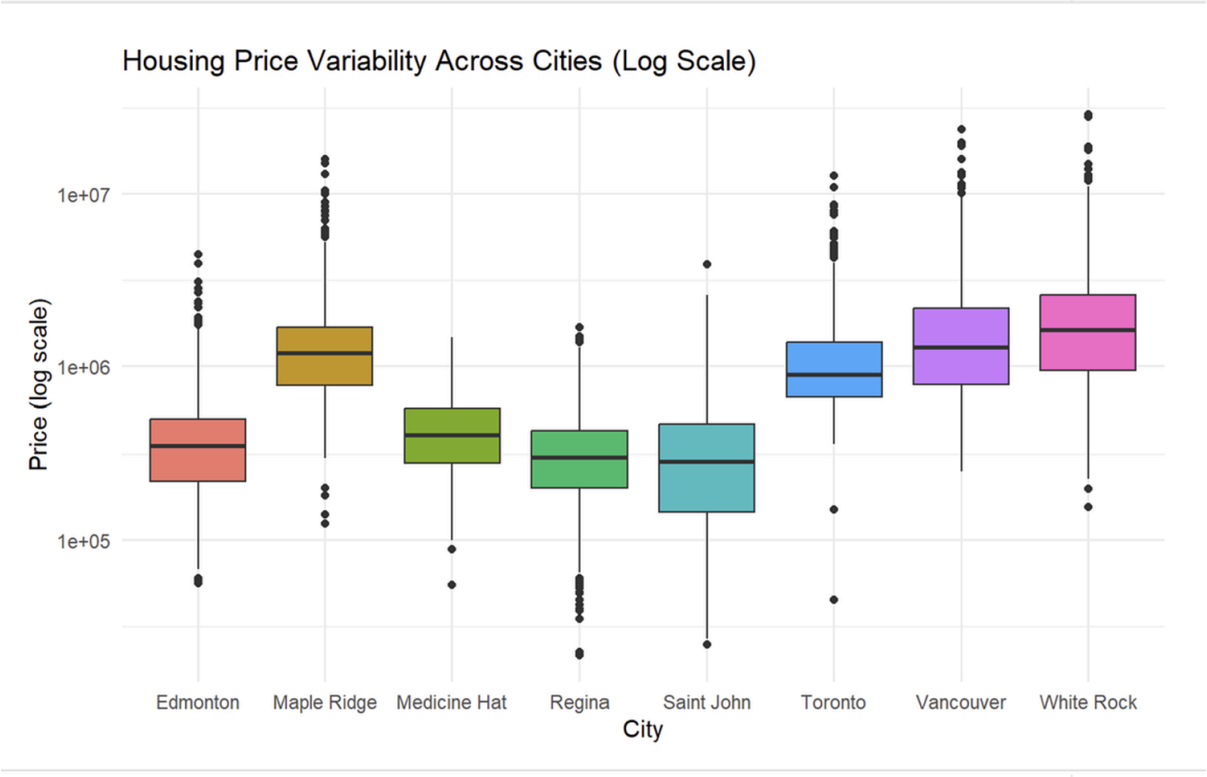

Figure 4. Box Plot of Housing Prices Across some Cities

Overall, the effectiveness of every sampling method is heavily reliant on the structure's population. Since we experienced significant variation between the housing prices across the forty-five most populated cities in Canada, this made stratification the ideal sampling strategy for obtaining achievable and precise estimates.

Conclusions and Discussion

The results above showed how important it is when choosing an appropriate sampling design based on the structure of the population. Housing prices create strong heterogeneity in the data, as they vary significantly. Methods like Stratified Sampling, which considered the variation, performed substantially better than those that did not.

The most efficient method is Stratified Sampling, especially with Neyman Allocation, as Neyman Allocation reduced the overall variance by assigning more samples to cities with higher variability. On the other hand, Proportional Allocation cannot fully capture differences in variability across strata, as it only considers stratum size.

Although Simple Random Sampling provided a reasonable baseline, it did not take the underlying structure of the data into consideration. Therefore, it has a higher variance

compared to Stratified Sampling. The Simple Random Sampling With Replacement reduced the efficiency further, as there are repeated observations, lowering the effective sample size.

As the cities are highly heterogeneous, Cluster Sampling also performed poorly, as the sample may not represent the full range of housing prices, as it only selected some subsets from the cities. The variability between cities still remains a major source of error, even through Two-Stage Cluster Sampling. This showed that Cluster Sampling is not a suitable method when clusters are internally homogeneous, but differ much from one another.

The above findings are consistent with the sampling theory that Stratification improves precision with strong between-group variation, but Cluster Sampling is less effective with that. However, this study has some limitations, such as the usage of a single dataset, so the results might not be applicable for other regions and types of data. Also, the high variability between cities affects Cluster Sampling performance.

In conclusion, this study compared different sampling methods for estimating the housing prices across Canadian cities, and the results showed that Stratified Sampling, especially with Neyman Allocation, has the most accurate estimates. Simple Random Sampling performs worse than the Stratification, but still better than the Cluster Sampling with high variability between cities. These findings suggest that considering the population's heterogeneity is essential when designing sampling strategies. To be practical, Stratified Sampling is preferred for data with clear group differences, and Neyman Allocation should be used to improve efficiency.

References

https://www.kaggle.com/datasets/jeremylarcher/canadian-house-prices-for-top-cities

Wu, C., & Thompson, M. E. (2020). *Sampling theory and practice*. Springer.